Annex IV. Debt Sustainability Analysis[1]

The Kyrgyz Republic remains at a moderate risk of debt distress, but massive foreign financed public investment in combination with the recent sharp depreciation of the currency is undermining debt sustainability. While under the DSA, external debt is estimated at 35 percent of GDP and remittances[2] in present value (PV) term, additional borrowing or further exchange rate volatility could take debt above the risk of debt distress. Against this background, the authorities need to exercise more prudence when contracting and guaranteeing new debt. Furthermore, a primary balance of about 2 percent of GDP by 2018 would be needed in order to bring public debt to sustainable levels. The authorities broadly concurred with the results of this DSA.

1. The 2015 updated joint Bank-Fund debt sustainability analysis (DSA) confirms the moderate risk rating found in the last DSA conducted last May (country report No.15/113). The update takes into account revised macroeconomic assumptions and updated data on public investment programs (PIP) borrowings.

Underlying DSA assumptions

2. This DSA is consistent with the macroeconomic framework outlined in the Staff Report for the First Review under the ECF (Box 1). For 2015, economic growth is expected to be slightly higher due to stronger than expected resilience of domestic demand to adverse external shocks. The fiscal balance in 2015 is expected to improve due to higher than expected nontax revenue and delays in the implementation of some infrastructure projects. The current account deficit has widened due to a larger than expected drop in remittances at the beginning of the year and the reclassification of the Russian-Kyrgyz Development Fund (RKDF) from current transfers to FDI due to its status of an international organization.

[1] Prepared by the staffs of the International Monetary Fund and the International Development Association. Approved by Juha Kähkönen and Masato Miyazaki (IMF) and Satu Kähkönen (IDA).

[2] The Kyrgyz Republic is classified as a large remittances case. In the 2012–14 period remittances averaged 29 percent of GDP and 61 percent of exports of goods and services, well above the 10 percent and 20 percent respective threshold-criteria for the large remittances case.

Selected Indicators (2014–35)

	2014	2015	2016	2017	2018	Long term
	Act.	Proj.				
Real GDP growth						
Current DSA	3.6	2.4	3.6	3.1	5.3	5.2
Previous DSA	3.6	1.7	3.4	5.2	5.2	5.2
Overall fiscal balance (percent of GDP)						
Current DSA	-3.9	-5.4	-9.1	-4.1	-1.1	-2.5
Previous DSA	-3.7	-7.6	-6.5	-3.9	-1.3	-2.5
Current account balance (percent of GDP)						
Current DSA	-16.8	-16.0	-17.3	-14.9	-14.1	-7.9
Previous DSA	-13.7	-17.0	-15.1	-13.4	-10.6	-7.9

Sources: Authorities data and IMF staff calculations.

Box 1. Macroeconomic Assumptions for 2015–35

Real GDP growth: Following a slowdown due to inclement weather (in 2014) and weak regional growth (in 2015) real GDP growth is expected to recover gradually in the coming years and reach its potential estimated at 5 percent by 2018. Growth in the medium to long run will be supported by infrastructure and energy investments, strong private credit growth and policy reforms aimed at improving the business climate.

Inflation is expected to remain at single-digit levels due to low fuel and food prices and prudent monetary policy. In the long-run inflation is anticipated to decrease further to 4 percent.

The current account deficit is expected to remain above 16 percent of GDP in 2015–16, driven by lower gold exports, the import content of the large investment projects and the drop in remittances in the context of the regional slowdown. After the unwinding of the public investment projects, and driven by a recovery of remittances, the current account deficit should stabilize at around 10 percent of GDP in the long-run.

Public investment: While PIP disbursement were somewhat delayed in 2015 (US$344 million instead of the projected US$505 million), they are expected to pick up in the medium run. As a result, capital expenditure is expected to amount to 16.5 percent of GDP in 2016 exceeding the level of private investment. Public investment should gradually decline beyond 2020 to stabilize at around 8 percent of GDP.

FDI: The reclassification of the Russian-Kyrgyz Development Fund (RKDF) resulted in an upward revision to FDI levels in the short to medium run. The implementation of the main public infrastructure projects is expected to increase net FDI flows to 8 percent of GDP in 2016–17. In the long-run FDI should stabilize at 5 percent of GDP.

The overall fiscal deficit is expected to increase above 4 percent by 2016 driven by elevated capital expenditures owing to public infrastructure projects. Following the unwinding of large public investments, the fiscal balance should stabilize gradually at around 1 percent of GDP in the long-run.

Box 1. Macroeconomic Assumptions for 2015–35 (concluded)

Financing: FDI and concessional borrowing are expected to be the main source of financing for public infrastructure investments. The average grant element of new external borrowing is projected to remain above 40 percent in the medium-run, but in line with our assumption of a gradual shift towards market financing beyond 2020, effective interest rates are expected to increase from the current 2 percent to close to 7 percent by the end of the forecast period. Remittances are projected to remain a major source of financing (close to 28 percent of GDP in 2016–17), but their share in percent of GDP should decline to 20 percent over the long term reflecting the diversification of the economy

3. The baseline scenario indicates a deterioration of the external debt burden in comparison with the previous DSA. External public and publicly guaranteed debt is projected at 63 percent of GDP in 2015,[3] which is 3 percentage points above the previous DSA. The higher than expected debt level in 2015 is mainly due to the 16 percent depreciation of the som vis-à-vis the U.S. dollar, while the delays in PIP disbursements contained the accumulation of external debt. In the medium-run, strong public investment is expected to keep external PPG debt at elevated levels, close to 65 percent of GDP.

Comparison of Debt Ratio

	2014	2015	2016	2017	2018	2019
	Act.	Proj.				
PPGE debt to GDP ratio						
Current DSA	51.0	63.0	65.3	66.1	64.7	63.6
Previous DSA	51.0	57.0	59.5	59.5	56.4	53.7
Public debt to GDP ratio						
Current DSA	53.0	64.9	66.9	67.6	66.1	64.9
Previous DSA	53.0	58.8	61.2	61.0	57.7	54.9

4. Planned infrastructure investments and depreciation of the national currency since the beginning of the year have brought external PPG debt closer to the high risk of debt distress level threshold. The present value of the public and publicly guaranteed (PPG) debt to GDP and remittances ratio is expected to reach 35 percent in 2016 and 2017, above the level envisaged in the previous DSA and within a 10 percent band around the 36 percent (34.2–37.8) corresponding threshold, signaling the Kyrgyz Republic as a borderline case. As regards the other solvency and liquidity indicators they do not breach their indicative thresholds and imply limited risks.

5. The external PPG debt outlook also remains vulnerable to large shocks, in particular to a further depreciation of the exchange rate. The PV of the debt to GDP and remittances ratio rises above the relevant indicative thresholds over the medium term under four of the six standard

[3] External public and publicly guaranteed debt includes the debt of the 19 largest SOEs.

stress tests: (i) when the exports grow at one standard deviation below their historical average over 2016–17; (ii) when the net non-debt creating inflows over 2016–17 are one standard deviation below their historical average; (iii) when the exchange rate depreciates by 30 percent in 2016 relative to the baseline; and (iv) under a shock over 2016–17 combining lower GDP and export growth, a decline in the U.S. dollar GDP deflator and lower net nondebt creating inflows (see Table 2). The shock on the exchange rate, which in magnitude is not far from the actual depreciation experienced this year, in particular would lead to a significant and protracted breach of the indicative threshold over the medium term.

6. The sharp decline in remittances presents an additional downside risk. Under the baseline scenario we assume a 20 percent drop in remittances in USD terms driven by the regional economic slowdown in 2015, followed by a gradual recovery in 2016 and 2017. Under a customized alternative scenario, which assumes an additional one-time 10 percent drop in remittances, the present value of the debt to GDP ratio would remain above 37 percent for a prolonged period after 2017, further elevating debt levels towards high risk of debt distress. Moreover, the private external debt also increased with the exchange rate depreciation adding stress on the overall external debt sustainability.

7. The probability approach confirms the Kyrgyz Republic's external risk rating as moderate.[4] Given that one of the indicative thresholds is nearly breached under the baseline and several thresholds are breached under the stress test scenarios, Kyrgyz is classified as a moderate-high borderline case. The probability approach indicates no breaches of the probability thresholds (Figure 3), confirming the moderate risk of external debt distress.

Public debt sustainability

8. The public debt outlook has also deteriorated compared to the previous DSA, mainly driven by the increase in external public debt following the depreciation of the som. Public debt is expected to reach 64.9 percent of GDP in 2015 and increase further to 67.6 percent of GDP in 2017 following the disbursement for the public infrastructure projects.

9. Total public debt is expected to be manageable in the medium and long term. Under the baseline scenario, the present value of public debt to GDP ratio would reach 46.7 percent in 2017, followed by a decline in the context of medium term fiscal consolidation. In the longer run, the present value of total public debt is expected to stabilize at 45 percent of GDP. Alternative scenarios and stress tests show that public debt remains highly sensitive to shocks that reduce real GDP growth, weaken the exchange rate or deteriorate the primary deficit over the medium term.

[4] The probability approach draws on country specific CPIA and GDP growth information to project debt distress probabilities.

10. The recently prepared medium-term debt strategy will need to be updated. It needs to include the most recent macroeconomic developments and the revised medium-term macroeconomic framework. While limiting the foreign exchange exposure of the debt portfolio would help to lower exchange rate vulnerabilities, it would need to be weighed against potentially higher financing costs due to less concessional terms available on domestic markets. Furthermore, establishing a viable alternative to external borrowing might take considerable time given the lack of depth of local bond markets.

Conclusion

11. While external PPG debt is expected to remain at a moderate risk of debt distress, immediate action is needed to contain it and bring it back to sustainable levels. The medium-term fiscal framework has been revised and now contains consolidation efforts totaling 4.6 percent of GDP in the 2016–18 period. Debt monitoring to minimize fiscal risk stemming from SOEs should be strengthened. Debt management should also benefit from the review of the public investment framework that authorities committed to undertake.[5]

12. The Kyrgyz authorities broadly concurred with the assumptions and conclusions, including identified vulnerabilities, of this DSA. They agreed that debt vulnerabilities have increased since the beginning of the year bringing the country to the borderline of high risk of debt distress. The authorities expressed their will to maintain public external debt at sustainable levels over the medium-term. To help them meet this objective, and given elevated debt vulnerabilities, staff agreed with the authorities' request to maintain an indicative target on zero-non-concessional borrowing. The authorities are also committed to a primary balance of about 2 percent of GDP by 2018 to ensure debt sustainability.

[5] The World Bank has provided technical assistance to the authorities in recent years on developing a medium-term debt management strategy, focusing on capacity building and on the quantification of costs and risks of alternative borrowing strategies, which lead eventually to a structured approach to future borrowing.

External Debt Sustainability Framework, Baseline Scenario, 2015–35[1/]

	Actual			Historical[6/] Average	Standard[6/] Deviation	Projections									
	2012	2013	2014			2015	2016	2017	2018	2019	2020	2015-2020 Average	2025	2035	2021-2035 Average
External debt (nominal) 1/	**80.2**	**72.9**	**81.2**			**98.1**	**100.8**	**100.7**	**97.9**	**95.8**	**94.9**		**86.0**	**76.0**	
of which: public and publicly guaranteed (PPG)	46.3	43.7	51.0			63.0	65.3	66.1	64.7	63.6	62.7		50.8	35.8	
Change in external debt	3.3	-7.3	8.3			16.9	2.7	-0.1	-2.8	-2.1	-0.9		-1.5	-1.8	
Identified net debt-creating flows	6.5	-1.5	13.8			6.1	5.2	5.0	2.4	1.5	-0.5		-2.3	-4.0	
Non-interest current account deficit	**14.9**	**14.1**	**15.9**	**7.9**	**6.9**	**14.6**	**15.6**	**13.1**	**12.8**	**11.3**	**8.6**	**12.6**	**3.8**	**-0.5**	2.8
Deficit in balance of goods and services	44.3	40.4	44.1			38.4	41.9	41.5	40.2	36.4	33.0		26.4	18.3	
Exports	49.0	51.3	44.0			41.1	43.0	40.8	40.0	39.7	40.2		42.6	47.3	
Imports	93.2	91.7	88.1			79.5	84.9	82.4	80.2	76.1	73.2		69.0	65.6	
Net current transfers (negative = inflow)	-31.2	-31.1	-30.4	-27.9	3.3	-26.0	-27.9	-29.9	-28.8	-27.9	-26.8	**-27.9**	-24.7	-20.3	-23.4
of which: official	-1.0	-1.7	-2.8			-1.5	-0.6	-1.3	-0.9	-0.6	0.0		0.0	0.0	
Other current account flows (negative = net inflow)	1.9	4.8	2.1			2.2	1.6	1.4	1.4	2.7	2.4		2.0	1.6	
Net FDI (negative = inflow)	**-4.4**	**-8.5**	**-2.3**	**-6.1**	**3.1**	**-7.8**	**-8.7**	**-7.1**	**-6.8**	**-6.9**	**-6.6**	**-7.3**	**-5.3**	**-4.8**	-5.3
Endogenous debt dynamics 2/	**-4.0**	**-7.0**	**0.3**			**-0.6**	**-1.8**	**-1.1**	**-3.6**	**-2.8**	**-2.5**		**-0.8**	**1.2**	
Contribution from nominal interest rate	0.7	1.0	1.0			1.5	1.7	1.8	1.3	1.8	2.8		3.7	5.0	
Contribution from real GDP growth	0.6	-7.6	-2.6			-2.2	-3.5	-2.9	-4.9	-4.7	-5.3		-4.4	-3.7	
Contribution from price and exchange rate changes	-5.4	-0.4	1.9			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**-3.2**	**-5.8**	**-5.6**			**10.8**	**-2.4**	**-5.0**	**-5.3**	**-3.6**	**-0.4**		**0.8**	**2.2**	
of which: exceptional financing	0.0	0.0	0.0			0.0	-0.4	-0.4	-0.4	-0.3	-0.3		-0.2	0.0	
PV of external debt 4/	...	...	65.8			78.2	80.1	79.8	77.3	75.5	75.1		70.5	64.4	
In percent of exports	...	...	149.6			190.1	186.3	195.6	193.4	190.4	186.9		165.4	136.3	
PV of PPG external debt	**...**	**...**	**35.6**			**43.2**	**44.5**	**45.2**	**44.2**	**43.4**	**42.8**		**35.2**	**24.3**	
In percent of exports	**...**	**...**	**81.0**			**105.0**	**103.6**	**110.6**	**110.5**	**109.4**	**106.6**		**82.7**	**51.4**	
In percent of government revenues	**...**	**...**	**110.2**			**130.5**	**139.3**	**142.1**	**136.2**	**131.3**	**127.6**		**105.2**	**59.3**	
Debt service-to-exports ratio (in percent)	**13.2**	**19.9**	**14.8**			**19.0**	**21.8**	**23.4**	**23.6**	**25.4**	**28.1**		**36.7**	**48.7**	
PPG debt service-to-exports ratio (in percent)	**4.5**	**7.7**	**3.6**			**4.9**	**6.2**	**6.6**	**6.9**	**7.1**	**6.9**		**6.8**	**6.1**	
PPG debt service-to-revenue ratio (in percent)	**6.9**	**12.3**	**4.9**			**6.1**	**8.4**	**8.5**	**8.6**	**8.5**	**8.2**		**8.6**	**7.1**	
Total gross financing need (Billions of U.S. dollars)	1.1	1.2	1.5			1.0	1.1	1.1	1.2	1.2	1.2		1.9	4.6	
Non-interest current account deficit that stabilizes debt ratio	11.6	21.4	7.6			-2.4	12.9	13.2	15.6	13.4	9.4		5.3	1.3	
Key macroeconomic assumptions															
Real GDP growth (in percent)	-0.9	10.5	3.6	4.1	4.0	2.4	3.6	3.1	5.3	5.2	6.0	4.3	5.5	5.1	5.0
GDP deflator in US dollar terms (change in percent)	7.5	0.4	-2.6	9.1	12.1	-12.1	-2.2	4.8	4.0	2.9	2.0	-0.1	2.0	2.0	2.0
Effective interest rate (percent) 5/	1.0	1.3	1.4	1.2	0.3	1.7	1.8	2.0	1.4	2.0	3.2	2.0	4.5	6.9	5.4
Growth of exports of G&S (US dollar terms, in percent)	3.4	16.3	-13.5	13.8	19.8	-15.8	5.8	2.7	7.3	7.3	9.5	2.8	8.7	8.1	8.3
Growth of imports of G&S (US dollar terms, in percent)	25.7	9.2	-3.0	21.6	25.1	-18.8	8.2	4.9	6.7	2.7	4.0	1.3	6.9	6.6	6.3
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	40.8	39.3	38.7	42.0	42.4	45.1	41.4	39.6	27.7	35.6
Government revenues (excluding grants, in percent of GDP)	32.4	32.0	32.3			33.1	32.0	31.8	32.4	33.0	33.6		33.5	41.0	34.4
Aid flows (in Billions of US dollars) 7/	0.3	0.3	0.3			0.4	0.5	0.4	0.5	0.5	0.5		0.5	0.7	
of which: Grants	0.2	0.2	0.3			0.3	0.3	0.3	0.2	0.2	0.2		0.2	0.3	
of which: Concessional loans	0.1	0.1	0.1			0.2	0.2	0.2	0.3	0.3	0.4		0.3	0.4	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			6.7	7.6	6.5	5.6	4.9	3.9		2.8	2.0	2.5
Grant-equivalent financing (in percent of external financing) 8/	0.0	0.0	0.0			65.9	56.2	58.6	60.1	59.5	59.6		57.7	43.2	53.3
Memorandum items:															
Nominal GDP (Billions of US dollars)	6.6	7.3	7.4			6.7	6.7	7.3	8.0	8.6	9.3		13.2	26.1	
Nominal dollar GDP growth	6.6	11.0	0.9			-10.0	1.3	8.1	9.6	8.2	8.1	4.2	7.5	7.2	7.1
PV of PPG external debt (in Billions of US dollars)			2.4			2.6	3.0	3.3	3.5	3.7	4.0		4.6	6.3	
(PVt-PVt-1)/GDPt-1 (in percent)						3.0	5.3	4.3	3.2	2.7	2.9	3.6	1.2	1.0	1.0
Gross workers' remittances (Billions of US dollars)	2.0	2.2	2.0			1.6	1.8	2.1	2.2	2.4	2.5		3.3	5.3	
PV of PPG external debt (in percent of GDP + remittances)	...	...	27.9			34.7	34.9	35.1	34.5	34.1	33.8		28.3	20.2	
PV of PPG external debt (in percent of exports + remittances)	...	...	49.8			65.8	63.3	65.1	65.1	64.7	63.9		52.3	36.0	
Debt service of PPG external debt (in percent of exports + remittance	...	...	2.2			3.1	3.8	3.9	4.1	4.2	4.1		4.3	4.3	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

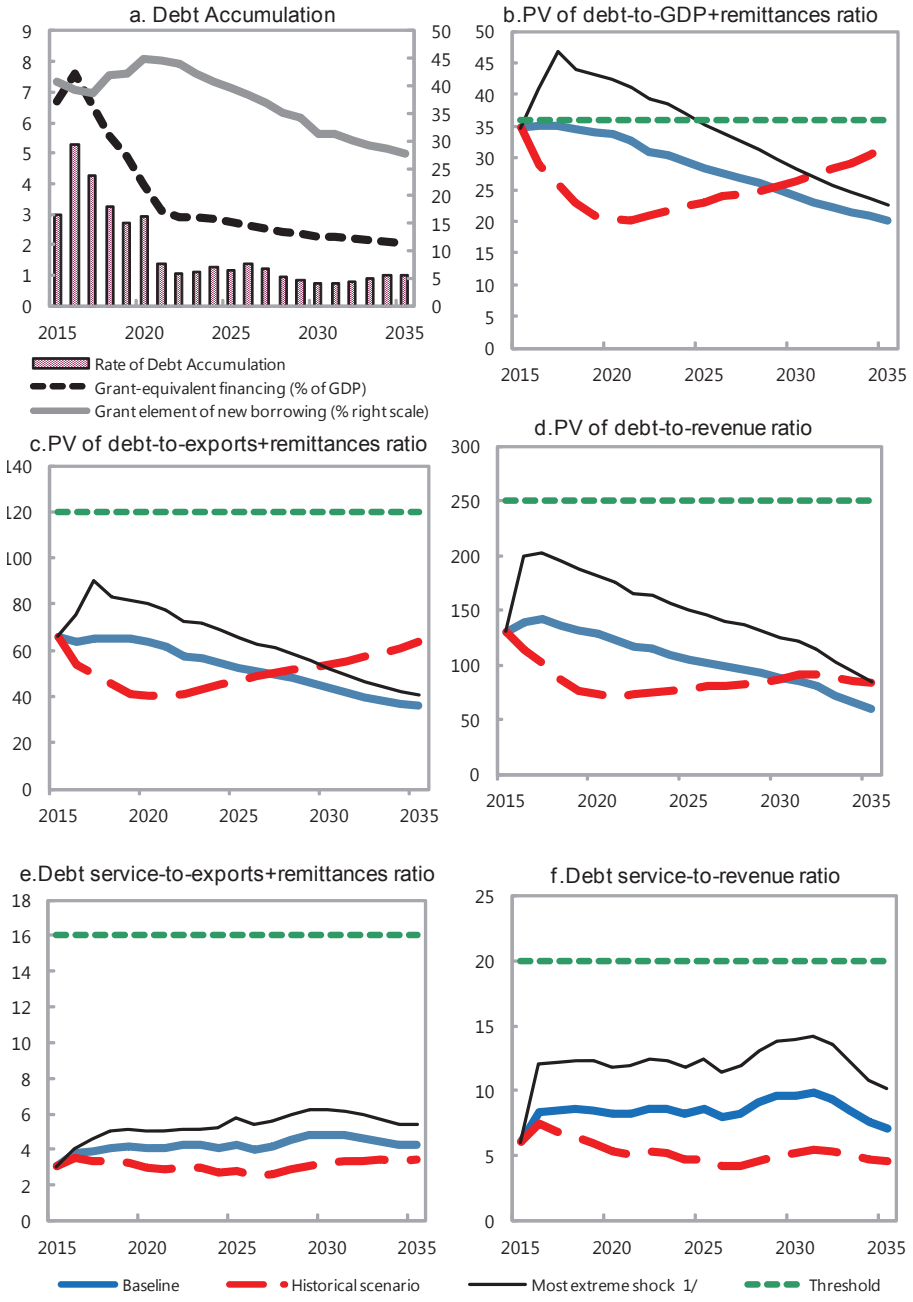
5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

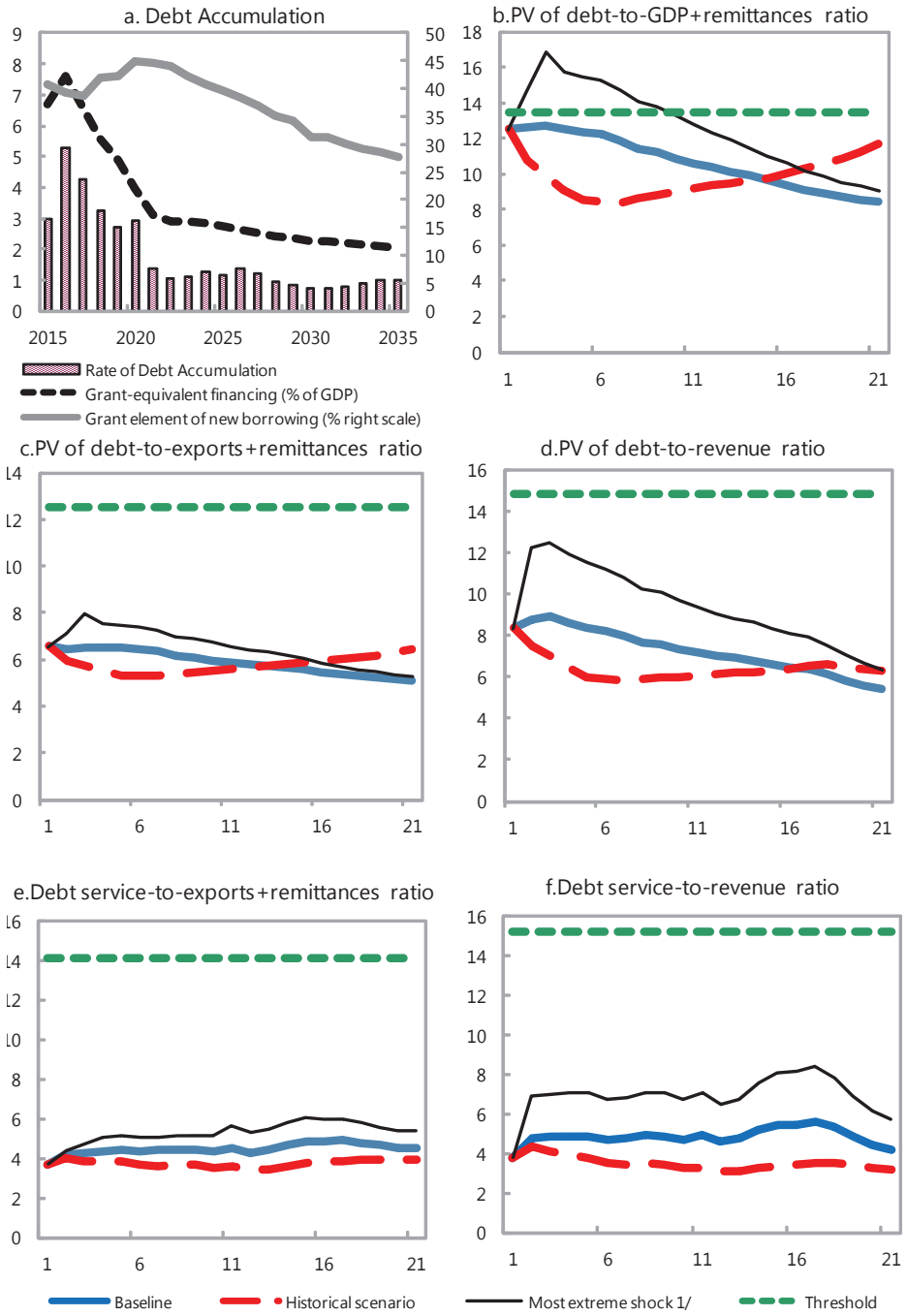
Indicators of Public and Publicly Guaranteed External Debt Under Alternatives Scenarios, 2015–35[1/]



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2025. In figure b. it corresponds to a Critical threshold shock; in c. to a Non-debt flows shock; in d. to a One-time depreciation shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

Probability of Debt Distress of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2015–35[1/]



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2025. In figure b. it corresponds to a Critical threshold shock; in c. to a Non-debt flows shock; in d. to a One-time depreciation shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2015–35

	Projections							
	2015	2016	2017	2018	2019	2020	**2025**	2035
PV of debt-to-GDP+remittances ratio								
Baseline	35	35	35	35	34	34	**28**	20
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	35	29	26	23	21	20	**23**	32
A2. New public sector loans on less favorable terms in 2015-2035 2	35	36	38	38	39	39	**36**	35
A3. Alternative Scenario : One time 10 percent drop in remittances	35	35	37	36	37	37	38	33
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	35	35	36	36	35	35	**29**	21
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	35	37	42	41	40	40	**33**	22
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	35	35	37	36	36	36	**30**	21
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	35	41	47	44	43	43	**35**	23
B5. Combination of B1-B4 using one-half standard deviation shocks	35	37	42	40	39	39	**32**	21
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	35	46	46	45	44	44	**37**	27
PV of debt-to-exports+remittances ratio								
Baseline	66	63	65	65	65	64	**52**	36
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	66	53	50	45	41	40	**47**	63
A2. New public sector loans on less favorable terms in 2015-2035 2	66	66	70	73	74	75	**67**	63
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	66	63	64	64	64	63	**52**	35
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	66	73	87	86	86	84	**69**	45
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	66	63	64	64	64	63	**52**	35
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	66	76	90	83	82	80	**65**	40
B5. Combination of B1-B4 using one-half standard deviation shocks	66	71	81	76	76	75	**61**	39
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	66	63	64	64	64	63	**52**	35
PV of debt-to-revenue ratio								
Baseline	131	139	142	136	131	128	**105**	59
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	131	113	102	87	76	73	**78**	83
A2. New public sector loans on less favorable terms in 2015-2035 2	131	145	154	152	150	149	**136**	103
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	131	143	150	144	138	135	**111**	62
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	131	149	169	161	155	150	**123**	64
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	131	139	153	147	141	137	**113**	63
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	131	158	182	173	166	161	**131**	67
B5. Combination of B1-B4 using one-half standard deviation shocks	131	145	165	157	151	146	**120**	63
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	131	199	203	194	187	182	**150**	84

Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2015–35 (continued)

(In percent)

Debt service-to-exports+remittances ratio								
Baseline	3	4	4	4	4	4	**4**	4
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	3	4	3	3	3	3	**3**	3
A2. New public sector loans on less favorable terms in 2015-2035 2	3	4	4	4	5	5	**5**	5
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	3	4	4	4	4	4	**4**	4
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	3	4	5	5	5	5	**6**	5
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	3	4	4	4	4	4	**4**	4
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	3	4	5	5	5	5	**5**	5
B5. Combination of B1-B4 using one-half standard deviation shocks	3	4	4	5	5	4	**5**	5
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	3	4	4	4	4	4	**4**	4
Debt service-to-revenue ratio								
Baseline	6	8	8	9	9	8	**9**	7
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	6	7	7	6	6	5	**5**	5
A2. New public sector loans on less favorable terms in 2015-2035 2	6	8	9	9	9	9	**9**	8
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	6	9	9	9	9	9	**9**	8
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	6	8	9	9	9	9	**10**	8
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	6	8	9	9	9	9	**9**	8
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	6	8	9	10	10	9	**11**	8
B5. Combination of B1-B4 using one-half standard deviation shocks	6	8	9	9	9	9	**10**	8
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	6	12	12	12	12	12	**12**	10
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	27	27	27	27	27	27	**27**	27

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Public Sector Debt Sustainability Framework, Baseline Scenario, 2012–35

(In percent of GDP unless otherwise indicated)

	Actual					Estimate		Projections							
	2012	2013	2014	Average 5/	Standard Deviation 5/	2015	2016	2017	2018	2019	2020	2015-20 Average	2025	2035	2021-35 Average
Public sector debt 1/	49.0	46.1	53.0			64.9	66.9	67.6	66.1	64.9	63.8		64.5	56.5	
of which: foreign-currency denominated	46.9	43.7	51.0			63.0	65.3	66.1	64.7	63.6	62.7		50.8	35.8	
Change in public sector debt	-0.4	-2.9	6.9			11.9	2.0	0.7	-1.5	-1.2	-1.1		-0.6	-0.9	
Identified debt-creating flows	2.3	-1.0	6.8			11.0	2.8	-1.4	-5.2	-5.9	-6.4		-2.5	-1.7	
Primary deficit	4.4	2.7	3.0	2.7	2.0	4.5	8.2	3.1	0.2	-1.5	-2.2	2.0	1.3	1.1	1.2
Revenue and grants	34.9	34.4	35.9			37.4	35.7	35.4	35.3	35.5	35.3		34.9	42.0	
of which: grants	2.5	2.4	3.6			4.3	3.8	3.6	2.9	2.4	1.8		1.4	1.0	
Primary (noninterest) expenditure	39.3	37.2	38.9			41.8	43.9	38.5	35.5	34.0	33.1		36.2	43.1	
Automatic debt dynamics	-2.1	-3.8	3.8			6.5	-5.0	-4.1	-5.0	-4.1	-3.9		-3.5	-2.8	
Contribution from interest rate/growth differential	-0.3	-5.2	-2.0			-1.2	-2.6	-2.3	-3.7	-3.5	-3.9		-3.5	-2.7	
of which: contribution from average real interest rate	-0.7	-0.5	-0.4			0.0	-0.3	-0.3	-0.3	-0.3	-0.2		-0.1	0.1	
of which: contribution from real GDP growth	0.4	-4.7	-1.6			-1.2	-2.3	-2.0	-3.4	-3.2	-3.7		-3.4	-2.8	
Contribution from real exchange rate depreciation	-1.8	1.4	5.7			7.7	-2.4	-1.8	-1.3	-0.6	0.0		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	-0.4	-0.4	-0.4	-0.3	-0.3		-0.2	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	-0.4	-0.4	-0.4	-0.3	-0.3		-0.2	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-2.7	-1.8	0.2			0.9	-0.8	2.0	3.7	4.7	5.4		1.8	0.8	
Other Sustainability Indicators															
PV of public sector debt	...	...	37.7			45.0	46.2	46.7	45.5	44.6	43.9		48.9	45.0	
of which: foreign-currency denominated	...	...	35.6			43.2	44.5	45.2	44.2	43.4	42.8		35.2	24.3	
of which: external	...	...	35.6			43.2	44.5	45.2	44.2	43.4	42.8		35.2	24.3	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	8.2	7.7	6.3			7.7	11.9	6.7	3.7	2.0	1.2		11.6	6.5	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	104.8			120.5	129.2	131.9	128.9	125.7	124.4		139.9	107.1	
PV of public sector debt-to-revenue ratio (in percent)	...	...	116.5			136.2	144.5	146.9	140.4	135.0	130.9		145.8	109.7	
of which: external 3/	...	...	110.2			130.5	139.3	142.1	136.2	131.3	127.6		105.2	59.3	
Debt service-to-revenue and grants ratio (in percent) 4/	6.5	11.6	5.6			6.1	8.0	8.0	8.2	8.2	8.2		15.6	15.7	
Debt service-to-revenue ratio (in percent) 4/	7.0	12.5	6.3			6.9	8.9	8.9	8.9	8.8	8.6		16.3	16.1	
Primary deficit that stabilizes the debt-to-GDP ratio	4.8	5.6	-3.9			-7.4	6.2	2.5	1.6	-0.3	-1.1		1.9	2.0	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	-0.9	10.5	3.6	4.1	4.0	2.4	3.6	3.1	5.3	5.2	6.0	4.3	5.5	5.1	5.0
Average nominal interest rate on forex debt (in percent)	1.2	1.2	1.3	1.1	0.2	1.3	1.4	1.4	1.4	1.4	1.5	1.4	1.6	2.1	1.8
Average real interest rate on domestic debt (in percent)	5.5	16.7	6.0	1.6	8.1	7.8	2.4	1.9	1.9	1.9	1.9	3.0	0.0	0.0	0.0
Real exchange rate depreciation (in percent, + indicates depreciation	-3.9	3.4	13.8	-3.9	10.7	15.5	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	9.5	3.5	7.9	11.2	6.9	6.8	8.3	6.9	6.1	5.0	4.0	6.2	4.0	4.0	4.0
Growth of real primary spending (deflated by GDP deflator, in percer	6.3	4.6	8.5	2.0	3.2	10.0	8.8	-9.6	-3.0	0.7	3.4	1.7	4.9	12.4	7.0
Grant element of new external borrowing (in percent)	...	...	...	...	...	40.8	39.3	38.7	42.0	42.4	45.1	41.4	39.6	27.7	...

Sources: Country authorities; and staff estimates and projections.

1/ General government gross debt.

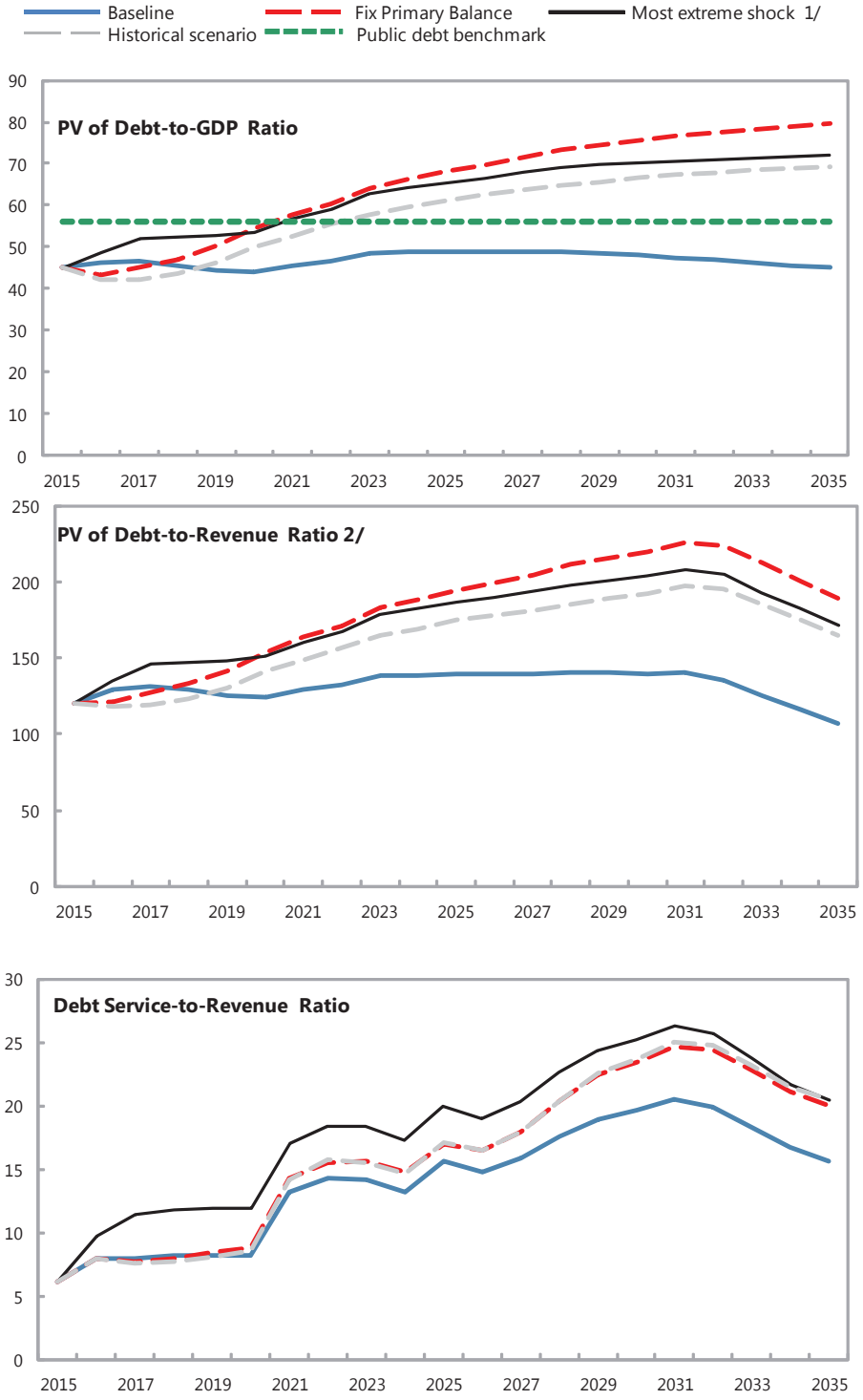
2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Indicators of Public Debt Under Alternative Scenarios, 2015–35[1/]



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2025.
2/ Revenues are defined inclusive of grants.

Sensitivity Analysis for Key Indicators of Public Debt 2015–35

	Projections							
	2015	2016	2017	2018	2019	2020	2025	2035
PV of Debt-to-GDP Ratio								
Baseline	45	46	47	46	45	44	49	45
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	45	42	42	44	46	50	61	69
A2. Primary balance is unchanged from 2015	45	43	45	47	50	54	68	79
A3. Permanently lower GDP growth 1/	45	47	48	48	48	49	64	94
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2016-20	45	49	52	52	53	54	65	72.1
B2. Primary balance is at historical average minus one standard deviations in 2016-201	45	44	46	44	44	43	48	44
B3. Combination of B1-B2 using one half standard deviation shocks	45	44	45	45	44	44	52	54
B4. One-time 30 percent real depreciation in 2016	45	63	61	58	56	55	58	52
B5. 10 percent of GDP increase in other debt-creating flows in 2016	45	53	54	52	51	50	54	48
PV of Debt-to-Revenue Ratio 2/								
Baseline	120	129	132	129	126	124	140	107
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	120	118	119	123	130	141	175	165
A2. Primary balance is unchanged from 2015	120	122	127	133	142	154	195	189
A3. Permanently lower GDP growth 1/	120	131	136	135	135	138	182	222
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2016-20	120	135	146	147	148	151	187	171
B2. Primary balance is at historical average minus one standard deviations in 2016-201	120	122	129	126	123	122	137	106
B3. Combination of B1-B2 using one half standard deviation shocks	120	122	127	126	125	126	150	127
B4. One-time 30 percent real depreciation in 2016	120	175	173	165	159	155	165	125
B5. 10 percent of GDP increase in other debt-creating flows in 2016	120	149	151	147	144	142	155	115
Debt Service-to-Revenue Ratio 2/								
Baseline	6	8	8	8	8	8	16	16
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	6	8	8	8	8	9	17	20
A2. Primary balance is unchanged from 2015	6	8	8	8	8	9	17	20
A3. Permanently lower GDP growth 1/	6	8	8	8	9	9	18	23
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2016-20	6	8	9	9	9	9	18	20
B2. Primary balance is at historical average minus one standard deviations in 2016-201	6	8	8	8	8	8	15	16
B3. Combination of B1-B2 using one half standard deviation shocks	6	8	8	8	8	8	16	17
B4. One-time 30 percent real depreciation in 2016	6	10	11	12	12	12	20	20
B5. 10 percent of GDP increase in other debt-creating flows in 2016	6	8	9	9	9	9	17	16

Sources: Country authorities; and staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.

2/ Revenues are defined inclusive of grants.